

Realizing an N-Type Organic Thermoelectric ZT of 0.46

Mingyu Ma, Gang Ye, Soyeong Jang, Yazhuo Kuang, Linlong Zhang, Shuyan Shao, L. Jan Anton Koster, Derya Baran, and Jian Liu*



Cite This: *ACS Energy Lett.* 2025, 10, 1813–1820



Read Online

ACCESS |



Metrics & More

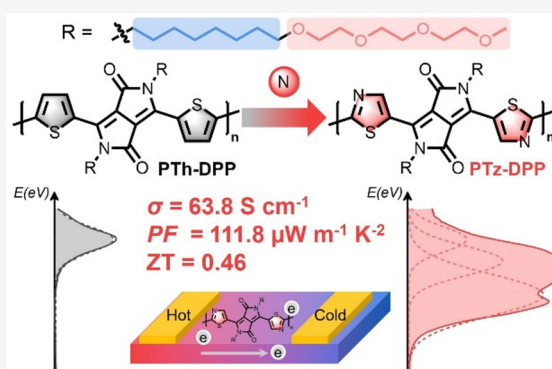


Article Recommendations



Supporting Information

ABSTRACT: The performance of n-type organic thermoelectric materials is significantly limited by a lack of a deep understanding of the structure–property relationship. In this study, we aim to establish a connection between the molecular structure and the density of states (DOS) profile related to thermoelectric performance. We synthesized three new diketopyrrolopyrrole-based polymers, each functionalized with amphipathic side chains. The only difference among these polymers is the number of sp^2 -N substitutions. Our findings indicate that as the number of substitutions increases, the DOS profile widens and intensifies, creating new peaks that extend toward the bandgap. This enables more efficient doping and coherent charge transport. Consequently, we achieved a high electrical conductivity of 63.8 S cm^{-1} , a power factor of $111.8 \text{ } \mu\text{W m}^{-1} \text{ K}^{-2}$, and a ZT of 0.46, representing a significant advancement in n-type organic thermoelectrics. This work provides valuable guidelines for designing high-performance thermoelectric materials by rationally tailoring the DOS profile.



Doped conjugated polymers are gaining tremendous attention as organic thermoelectric (OTE) materials due to their unique combination of low thermal conductivity, ease of processing, and favorable mechanical properties.^{1–7} They are ideal for creating portable, personal, and flexible thermoelectric modules.^{8,9} The performance of thermoelectric materials is measured by their figure-of-merit, $ZT = S^2\sigma/\kappa$.¹⁰ Here, S represents the Seebeck coefficient, σ represents the electrical conductivity, $S^2\sigma$ represents the power factor (PF), and κ represents the thermal conductivity. Pristine conjugated polymers are intrinsic semiconductors that lack free charge carriers. Molecular doping involves the addition of a (strong) molecular oxidizing or reducing agent, and it has been widely explored to optimize the electrical and thermal properties of conjugated polymers.^{11,12}

High-performing p-type and n-type OTE materials are necessary to create efficient thermoelectric modules. Recently, Di and colleagues reported a dimensionless thermoelectric figure of merit (ZT) of 0.74 at 298 K for a p-type polymeric multiheterojunction featuring periodic dual-heterojunctions. This represents a significant improvement over the ZT of 0.25 observed for the pure film of PDPPSe-12, indicating a promising future for organic thermoelectrics.^{13,14} However, the performance of n-type OTE materials is still significantly inferior to that of p-type counterparts.^{15–19} Many research

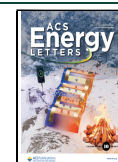
efforts have been made to improve n-type OTE materials by innovating electron-deficient building blocks to tune the energetics or improve backbone planarity.^{20–24} New side-chain designs, like branching point-varied alkyl types,^{25,26} ethylene glycol types,^{27,28} or amphipathic side chains,^{29,30} have been developed to optimize molecular packing, enhance host/dopant miscibility, or regulate the electrostatic interaction. There are various electron-deficient building blocks, and diketopyrrolopyrrole (DPP) is one of the most extensively studied for creating conjugated polymers of donor–acceptor (D-A) type for organic electronics.³¹ Most DPP-based D-A copolymers are either p-type or ambipolar, exhibiting a wide range of field-effect charge mobilities from 10^{-3} to $10 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$,^{32–34} but are challenging to be efficiently n-doped ($\sigma < 10^{-2} \text{ S cm}^{-1}$).³⁵ To tackle this challenge, many molecular strategies, such as using a twisted backbone, incorporating weak donor (or even acceptor) moieties, and flanked by nonthiophene units, have been adopted to tune the doping

Received: December 29, 2024

Revised: February 22, 2025

Accepted: March 14, 2025

Published: March 19, 2025



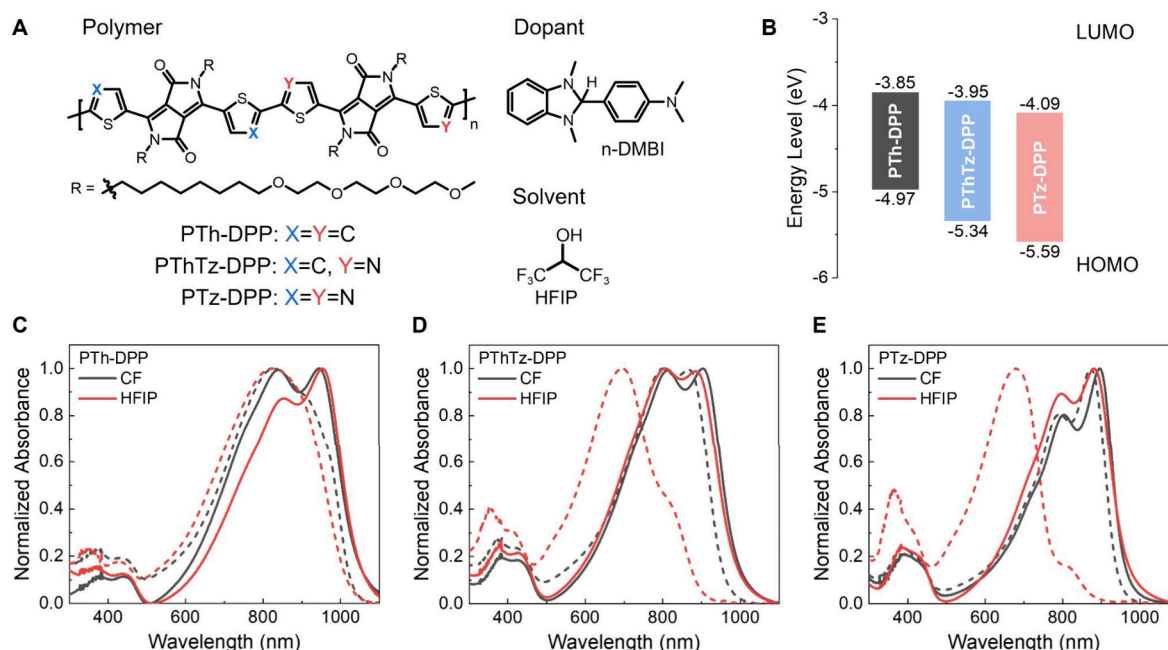


Figure 1. (A) Chemical structures of PTh-DPP, PThTz-DPP, PTz-DPP, n-DMBI, and processing solvents. (B) Energy levels of three polymers. (C–E) UV–vis–NIR absorption spectra of (C) PTh-DPP, (D) PThTz-DPP, and (E) PTz-DPP in solution and thin films. The dashed lines are the spectra in CF and HFIP solutions, and the solid lines are the spectra in films from CF and HFIP solution.

efficiency and thermoelectric performance.^{36–41} Despite significant progress, the ZT values of these n-type OTE polymers are less than 0.25, or are often not reported, with power factor values often under $100 \mu\text{W m}^{-1} \text{K}^{-2}$.

As the doping level gradually increases, σ and S of OTE materials obey an empirical power law of $S \propto \sigma^{-0.25}$, followed by a sharp decrease in S at high doping levels.^{42,43} The characteristic curvature of the S – σ relation is explained by the doping-induced broadening of the density of states (DOS) profile.^{44,45} So far, there are only a few studies on improving TE performance by tailoring the DOS profile. Liu et al. report that tailoring the DOS of a doped NDI-based polymer toward reducing the gap states can enhance the Seebeck coefficient without compromising electrical conductivity.⁴⁶ Pei and co-workers modified the DOS profile of a coplanar conjugated polymer by solvent engineering, optimizing charge transport.⁴⁷ Despite the significance of the DOS profile, there is limited knowledge about predicting DOS profiles from chemical structures. Hence, more efforts are required to set guidelines for reshaping the DOS profile through molecular approaches toward high-performance n-type OTEs.

In this contribution, we conducted a case study of n-type DPP-based polymers to explore the impact of sp^2 -nitrogen (N) substitution on the DOS profile and thermoelectric performance. Three new polymers, PTh-DPP, PThTz-DPP, and PTz-DPP, were synthesized by polymerizing thiophene-DPP and 3,6-di(thiazol-5-yl)-DPP, respectively, either separately or in combination. Amphipathic side chains were incorporated into the DPP units. The three polymers differ in the number of sp^2 -N substitutions in the repeating unit. As the number increases, the DOS profile widens and intensifies, forming new peaks that extend toward the bandgap. Consequently, the charge transport mechanism changes from a hopping-dominated mode to a coherent charge transport, leading to an enhanced electrical conductivity and power factor with the number of sp^2 -N substitutions. Among them, doped PTz-DPP has the

highest electrical conductivity value of 63.8 S cm^{-1} , an $S^2\sigma$ value of $111.8 \mu\text{W m}^{-1} \text{K}^{-2}$, and a figure-of-merit ZT of 0.46. This makes it one of the best options for n-type OTEs.

The chemical structures of polymers and the molecular dopant (n-DMBI) are shown in Figure 1A. The synthetic procedures and characterizations of three polymers are described in Supporting Information. All polymers have amphipathic side chain that enable good host/dopant miscibility and govern the electrostatics from the counterions upon molecular doping. According to the gel permeation chromatography with HFIP as the eluent at 35°C , the number-average molecular weight (M_n)/polydispersity index (PDI) of PTh-DPP, PThTz-DPP, and PTz-DPP were estimated to be 113 kDa/2.2, 122 kDa/2.2, and 137 kDa/2.5, respectively (Figure S2). PTh-DPP, PThTz-DPP, and PTz-DPP exhibit excellent thermal stability with decomposition temperatures (5% weight loss) of 382, 379, and 392°C , as revealed by thermogravimetric analysis (TGA) (Figure S3). The energy levels of the three polymers were determined via cyclic voltammograms (CV) (Figure 1B and Figure S4). The LUMO/HOMO energy levels for PTh-DPP, PThTz-DPP, and PTz-DPP are $-3.85/-4.97$, $-3.95/-5.34$, and $-4.09/-5.59$ eV, respectively. The deepening of the energy level as sp^2 -N substitution was validated by density functional theory (DFT) calculations (Figures S5 and S6). All three hexamers have a planar and quasilinear conjugated backbone.

Figure 1C–E demonstrates the normalized UV–vis–NIR absorption spectra for three intrinsic polymers in dilute chloroform (CF), hexafluoroisopropanol (HFIP) solution, and thin film states. The three DPP-based polymers exhibit typical two absorption peaks ($\lambda_{\text{max}} = 829 \text{ nm}$ for PTh-DPP, $\lambda_{\text{max}} = 800$ and 867 nm for PThTz-DPP, and $\lambda_{\text{max}} = 794$ and 873 nm for PTz-DPP) in dilute CF solution. When they transitioned from the CF solution to the solid state, the polymers showed an absorption maximum that shifted bathochromically by less than 20 nm. This suggests that the

polymers have an excellent backbone planarity.⁴⁸ They likely preaggregate in dilute chloroform due to a strong π – π interaction. The absorption profile of PTh-DPP remains primarily unchanged when changing from a dilute chloroform solution to a dilute HFIP solution. However, for PThTz-DPP and PTz-DPP there is a significant blue shift of λ_{max} by 110 and 118 nm, respectively. It appears that a polymer with more thiazole units shows stronger solvatochromism. As the HFIP solvent evaporates, such a solvation effect disappears, leading to similar π – π stacking characteristics and absorption spectra of the polymers in the solid state.

The n-doping of the three DPP-based polymers was performed by using 4-(1,3-dimethyl-2,3-dihydro-1H-benzomiazol-2-yl)-N,N-dimethylaniline (n-DMBI) as the n-type dopant, which proved effective.^{29,30} Figure 2A shows the

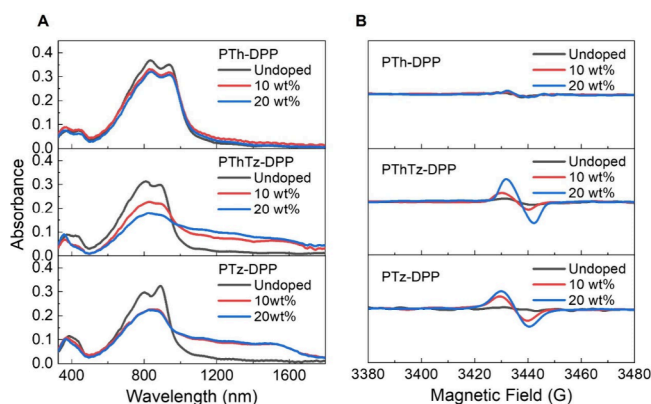


Figure 2. (A) UV–vis–NIR absorption spectra of PTh-DPP, PThTz-DPP, and PTz-DPP films at different doping ratios. (B) EPR spectra of PTh-DPP, PThTz-DPP, and PTz-DPP films at different doping concentrations.

doping-induced changes in the UV–vis–NIR absorption spectra of three types of polymer films. The doping process caused a decrease in low-energy neutral absorption bands that peaked at around 800 nm, while there was an increase in polaron absorption between 1000 and 1800 nm. Out of the three types of polymers, it was observed that the PTh-DPP film exhibited the least quenching of the neutral band. This indicates a weaker redox reaction with the dopant. On the other hand, the doped PThTz-DPP and PTz-DPP films showed stronger polaron absorption bands. In Figure 2B, electron paramagnetic resonance (EPR) spectroscopies indicated that all neat polymers possess weak radical signals, validating their open-shell structures.⁴⁹ When the doping concentration increases, the spin densities of the doped PThTz-DPP and PTz-DPP increase more significantly than those of the doped PTh-DPP. These results indicate that molecular doping generates more polarons as the number of sp^2 -N substitutions increases.

To investigate the molecular packing of these solid films, we carried out 2D grazing incidence wide-angle X-ray scattering (GIWAXS) measurements (Figures 3A, S7, and S8). The results showed that the PTh-DPP polymer primarily adopts a face-on orientation relative to the substrate, rendering the (100) peak at $q_{xy} = 0.244 \text{ \AA}^{-1}$ and the (010) peak at $q_z = 1.65 \text{ \AA}^{-1}$. These peaks indicate a lamellar packing with an interlayer spacing of 2.58 nm along the in-plane direction and a π – π stacking distance of 0.38 nm. As the number of sp^2 -N substitutions increases, the orientation of the chain remains

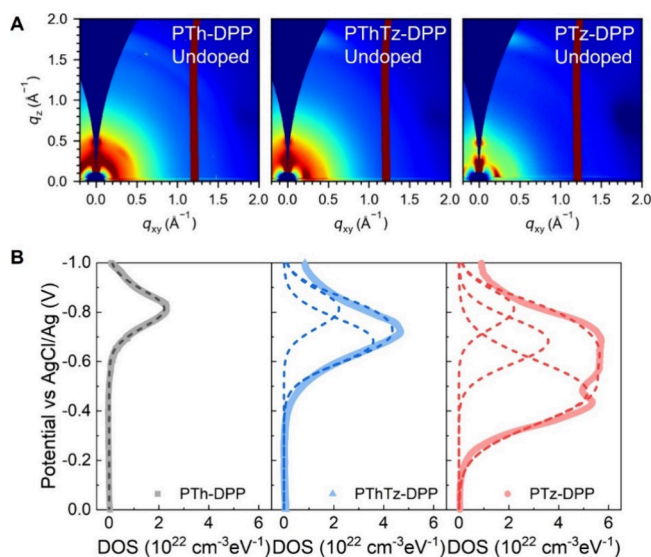


Figure 3. (A) 2D GIWAXS patterns of undoped PTh-DPP, PThTz-DPP, and PTz-DPP films. (B) DOS distributions of undoped PTh-DPP, PThTz-DPP, and PTz-DPP films. The dashed lines are Gaussian fits.

almost the same. However, the chain packs more closely with increased crystallinity in the direction of π – π stacking. The distance between the π -stacking of PThTz-DPP and PTz-DPP is 0.37 and 0.36 nm, respectively. This closer π -stacking of polymer chains results in better fibril-textured morphologies of PThTz-DPP and PTz-DPP compared with PTh-DPP (Figure S9). The closer π -stacking of the backbone may cause extended side chains, leading to a slightly increased lamellar spacing of 2.63 nm for PThTz-DPP and 2.64 nm for PTz-DPP. The three polymers are miscible with dopant molecules, likely due to their amphiphilic side chains.³⁰ After dopant molecules were incorporated, the previously well-organized arrangement of the PTh-DPP chain was significantly disrupted in both the π – π stacking and lamellar packing directions. Interestingly, PTz-DPP appears to be more resistant to molecular doping. As a result, the impact on the molecular order of chain packing is reduced.

To quantify the impact of substituting sp^2 -N on electronic structures of DPP-based polymers, we used an electrochemical method (Figure S10a–c) to measure the DOS distributions of pristine and 10 wt %-doped films.^{46,47,50,51} Figure 3B shows the DOS distributions for the three DPP-based polymer films versus the Ag/AgCl reference. To obtain the total density (N_m), center (E_{ct}), and disorder parameter (σ_d), the DOS distributions are fitted via the Gaussian function, and the results are summarized in Tables 1 and S1. For PTh-DPP, the DOS distribution can be described by a single Gaussian function. However, for PThTz-DPP, which contains half of Tz-DPP, a newly formed energy band appears in the DOS profile and requires fitting with two Gaussian functions. When Th-DPP is completely replaced with Tz-DPP, the total DOS distribution of PTz-DPP is further extended with a third peak. As the Tz-DPP content increases from PTh-DPP to PTz-DPP, the DOS profile widens and intensifies, forming new peaks that extend toward the bandgap. The disorder parameters of 77, 90, and 128 meV are obtained for the pristine films of PTh-DPP, PThTz-DPP, and PTz-DPP, respectively. The total densities of 4.3×10^{21} , 8.1×10^{21} , and $1.66 \times 10^{22} \text{ cm}^{-3}$ are obtained for the pristine films of PTh-DPP, PThTz-DPP, and PTz-DPP,

Table 1. Summary of the Thermoelectric Performance, Charge Mobility, and Total Density of the Three Polymers

polymer	σ^a (S cm ⁻¹)	S (μ V K ⁻¹)	PF (μ W m ⁻¹ K ⁻²)	κ (W m ⁻¹ K ⁻¹)	ZT	μ^b (cm ² V ⁻¹ s ⁻¹)	N_m^c (cm ⁻³)
PTh-DPP	0.03	-403.7	0.5	0.146		4.2×10^{-4}	4.3×10^{21}
PThTz-DPP	0.68	-277.0	5.2	0.176		0.13	8.1×10^{21}
PTz-DPP	63.2	-133.0	111.8	0.073	0.46	0.43	1.66×10^{22}

^aThese polymers' thermoelectric performance at optimal dopant concentration. ^bThe electron mobility from MIS devices at 10 wt %-doped. ^cThe total density (N_m) of undoped polymer films, which were calculated by Gaussian fitting.

respectively. The center of the DOS distribution in the pristine films of PTh-DPP, PThTz-DPP, and PTz-DPP is -0.81, -0.68, and -0.48 eV, respectively. The center position remains unchanged upon doping (Figure S11). The PTz-DPP film exhibits the deepest center and sizable total site density, facilitating doping for generating free-charge carriers with high mobility.

Figure 4A shows the electrical conductivity (σ) of the doped and undoped PTh-DPP, PThTz-DPP, and PTz-DPP thin films

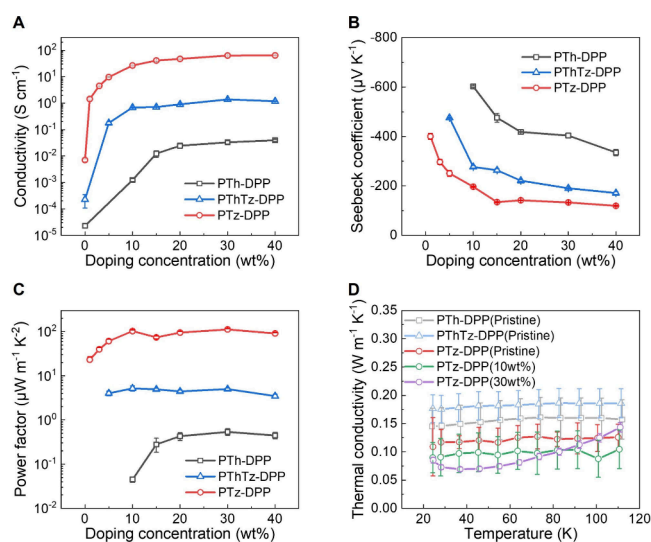


Figure 4. (A) Electrical conductivity, (B) Seebeck coefficient, (C) power factor, and (D) thermal conductivities of various doped PTh-DPP, PThTz-DPP, and PTz-DPP films.

at different doping concentrations (see details in Figures S13 and S14 in Supporting Information). The corresponding thermoelectric parameters are summarized in Table 1. The undoped PTh-DPP, PThTz-DPP, and PTz-DPP thin films have σ of 2.3×10^{-5} S cm⁻¹, 2.3×10^{-4} S cm⁻¹, and 7.1×10^{-3} S cm⁻¹, respectively. These values are higher than those ($<10^{-7}$ S cm⁻¹) reported for typical neat conjugated polymers, which are consistent with the nature of open-shell structures.⁵² The PTh-DPP and PThTz-DPP show the highest σ of 0.04 S cm⁻¹ and 1.38 S cm⁻¹, respectively. Meanwhile, the PTz-DPP polymer shows an optimized σ value of 63.8 S cm⁻¹ when the doping ratio reached 40 wt %. These results suggest that tailoring the DOS profile through backbone chemistry effectively enhances the electrical conductivity of conjugated polymers. Interestingly, the electrical conductivity of doped PTz-DPP films keeps growing even up to a high doping loading of 40 wt %. It suggests that the PTz-DPP polymer is tolerant of molecular doping^{23,49} or that the adverse electron–electron repulsion at high doping levels is overcome because of the unique molecular motif.⁵³

Figure 4B displays the Seebeck coefficients of various doped polymers (details in Figure S15). By adjusting the doping concentration from 10 to 40 wt %, the Seebeck coefficient of the doped PTh-DPP films changes from -602.5 ± 3.5 to -335.2 ± 13.8 μ V K⁻¹. As the number of sp²-N substitutions increases, the doped polymer films exhibit a lower absolute Seebeck coefficient at a given doping concentration. The doped PTz-DPP film exhibits a Seebeck coefficient from 400.3 ± 11.7 μ V K⁻¹ to -119.4 ± 1.4 μ V K⁻¹ by adjusting the doping concentration from 1 to 40 wt %. This indicates an apparent increase in doping levels from PTh-DPP to PThTz-DPP and PTz-DPP, which correlates well with the evolution of DOS profiles. The optimal power factors of PTh-DPP and PThTz-DPP are 0.5 ± 0.1 and 5.2 ± 0.7 μ W m⁻¹ K⁻² (Figure 4C), respectively. At a doping ratio of 30 wt %, the PTz-DPP film exhibits an optimized PF value of 111.8 ± 5.7 μ W m⁻¹ K⁻², representing one of the best values among n-type organic thermoelectrics.

Figure 4D displays three polymer films' variable temperature in-plane thermal conductivities, measured by the standard 3-omega (3 ω) method.⁵⁴ The pristine PTh-DPP and PThTz-DPP exhibit higher in-plane κ values of 0.146 ± 0.032 W m⁻¹ K⁻¹ and 0.176 ± 0.024 W m⁻¹ K⁻¹, respectively. The pristine PTz-DPP shows an in-plane κ value of 0.117 ± 0.022 W m⁻¹ K⁻¹ at room temperature (301 K), which is comparable to those reported values (≈ 0.13 W m⁻¹ K⁻¹) of p-type DPP-based conjugated polymers.¹⁴ As the doping concentration increases, the κ value slightly decreases, in accordance with a previous study.⁵⁵ At doping concentrations of 10 and 30 wt %, the κ values are 0.091 ± 0.033 W m⁻¹ K⁻¹ and 0.073 ± 0.004 W m⁻¹ K⁻¹, respectively. The high power factor and low thermal conductivity support the utilization of PTz-DPP as a highly efficient and promising n-type OTE material, giving a figure-of-merit ZT of 0.46.

The temperature dependence of electrical conductivity was analyzed by fitting it with the Arrhenius eq (Figures 5A and S14d). The activation energy values derived from the analysis are 289/249, 200/169, and 56/43 meV for 10/30 wt %-doped PTh-DPP, PThTz-DPP, and PTz-DPP films, respectively. As the number of sp²-N substitutions increases, the increased doping levels may contribute to the reduced activation energy. Besides, the intensified DOS distribution likely leads to a more coherent charge transport occurring in the vicinity of the Fermi level, which is supported by the small activation energy of the doped PTz-DPP within $2kT$. This charge transport process relies on the coherence of the wave function over distances of at least a few nanometers, resulting in band-like charge transport properties that produce pronounced Hall effect signals.⁵⁶

Not every radical species, such as a polaron or bipolaron, residing on the polymer backbone represents a free-charge carrier. To obtain the density of mobile charge carriers, we utilized admittance spectroscopy for ion gel-based metal–insulator–semiconductor (MIS) devices developed in our

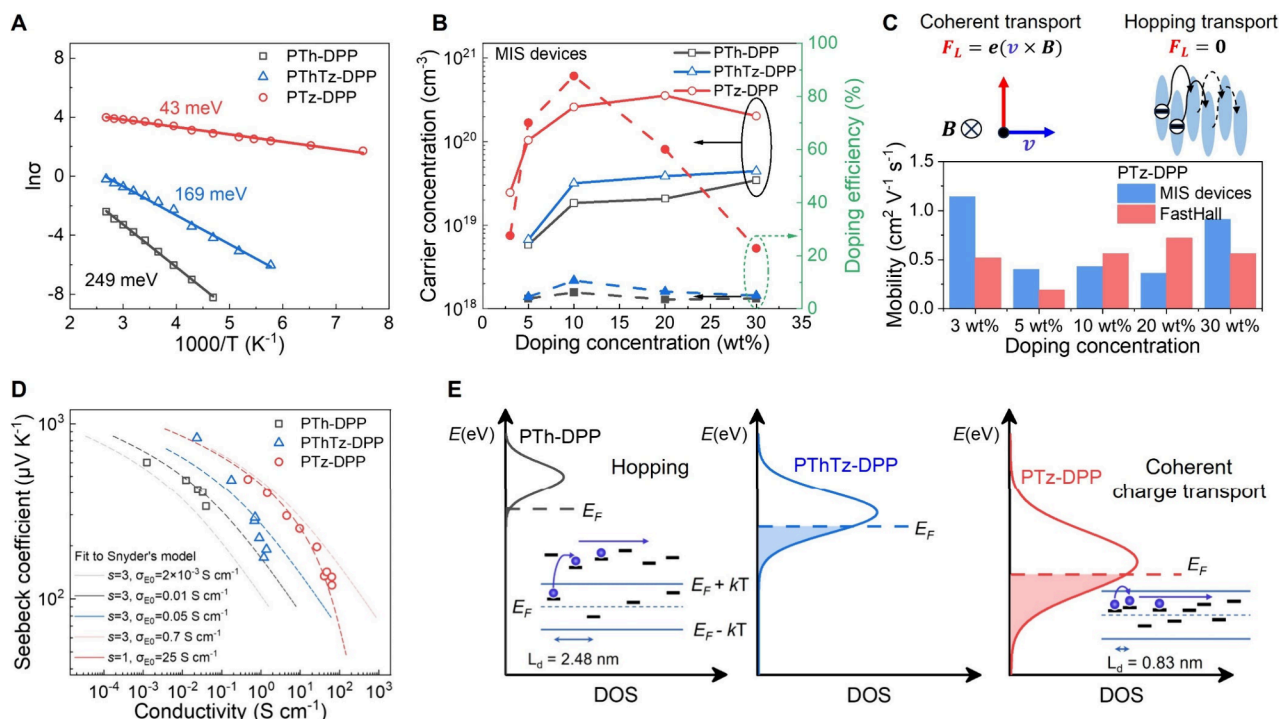


Figure 5. (A) Temperature dependence of the electrical conductivity of the 30 wt %-doped PTh-DPP, PThTz-DPP, and PTz-DPP films. (B) Carrier concentration and doping efficiency of PTh-DPP, PThTz-DPP, and PTz-DPP films based on the Mott–Schottky analysis. The doping efficiency is calculated from the equation $\eta = n/n_{\text{dopant}}$ where n is the carrier concentration and n_{dopant} is the n-DMBI density calculated from the molar ratio of dopant to the polymer repeating unit density. (C) Charge mobility of the Mott–Schottky analysis, and the FastHall measurement of PTz-DPP films. (D) Snyder's model fits with PTh-DPP, PThTz-DPP, and PTz-DPP. (E) DOS distributions and charge transport mechanisms of the polymers.

previous studies.^{28,30} As shown in Figure 5B, the carrier concentration of PTz-DPP is 1 order of magnitude higher than that of the other two polymers, and the peak doping efficiency approaches 80%, dictated mainly by the DOS profile. The large population of free charges contributes to the high electrical conductivity of the doped PTz-DPP.

The FastHall technique can address the Hall voltage compensation effect as hopping carriers respond to the transverse Hall electric field.^{57,58} It detects only charge carriers that adopt coherent charge transport (as illustrated in Figure 5C). The Hall-effect signal of the doped PTh-DPP film was within the noisy level, implying a hopping-dominated charge transport. On the contrary, the doped PTz-DPP films exhibit pronounced Hall-effect signals at the level of several μV, which are translated to charge mobilities ranging from 0.2 to 0.7 cm² V⁻¹ s⁻¹, which are comparable to those extracted from MIS devices (Figure 5C and Table 1). This suggests that a significant portion of charge carriers in doped PTz-DPP films follow a coherent charge transport mechanism.

We investigated the S – σ relationships of various doped polymers (Figure 5D). We almost fit the S – σ curve of doped PTh-DPP by using the Snyder's model⁴³ with a transport parameter of $s = 3$ and transport coefficient of $\sigma_{E0} = 0.01$ S cm⁻¹. Fitting to the S – σ curves of doped PThTz-DPP and doped PTz-DPP requires a higher transport coefficient. Interestingly, the S – σ relation of doped PTz-DPP appears to be better fitted with $s = 1$ and a high $\sigma_{E0} = 25$ S cm⁻¹. $s = 1$ is typically found in crystalline semiconductors and metals characterized by coherent charge transport.⁵⁹ These results indicate that by tailoring the DOS profile, the mechanism of charge moving has changed from a hopping transport for

doped PTh-DPP to a more coherent charge transport for doped PTz-DPP.

We estimated Fermi levels (E_F) of various doped polymers using their Seebeck coefficient values, as illustrated in Figure 5E. The sites with energy levels near E_F within the kT range can be determined by integrating the DOS function from $E_F - kT$ to $E_F + kT$. The number of these sites can be translated into the distance (L_d) between two adjacent sites within the energy band of $2kT$. The distance is 2.48, 1.05, and 0.83 nm, respectively, for 10 wt %-doped PTh-DPP, PThTz-DPP, and PTz-DPP films. The length (L_c) of the Th-DPP or Tz-DPP unit is approximately 1.1 nm, which dictates the conjugation length of the conjugated polymer. For the doped PTh-DPP film, the L_d is much larger than the length of the Th-DPP unit. It suggests that charges must hop to a nearer site with higher energy.⁶⁰ As for the doped PTz-DPP film, L_d is much shorter and within the length scale of the repeating unit. It may lead to coherent charge transport occurring near the Fermi level. This analysis is simplified but captures the essence of how the charge transport mechanism is affected by the DOS profile. Future molecular design should focus on maximizing the site density at energy levels that are distanced from the DOS maximum. This approach aims to achieve high charge mobilities with relatively low doping levels, resulting in enhanced power factors.

In summary, we demonstrated how sp²-N substitution influences the DOS profile and thermoelectric performance of n-type DPP-based polymers. As the number of sp²-N substitutions increases, the DOS profile widens and intensifies, resulting in new peaks that extend toward the bandgap. The tailoring of the DOS profile enhances the doping efficiency and

promotes more coherent charge transport, increasing the thermoelectric performance. The doped PTz-DPP has the highest electrical conductivity value of 63.8 S cm^{-1} , a power factor of $111.8 \mu\text{W m}^{-1} \text{ K}^{-2}$, and a figure-of-merit ZT of 0.46, making it one of the best n-type OTE performers. This study offers important guidelines for designing high-performing OTE materials by establishing DPP-based polymers and analyzing how heteroatom substitution affects the DOS profile, which, in turn, influences energy levels, doping efficiency, and transport properties.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsenerylett.4c03590>.

Experimental methods; Figure S1: Synthetic routes to polymers; Figure S2: GPC images; Figure S3: TGA images; Figure S4: CV images; Figures S5 and S6: DFT calculations; Figures S7 and S8: GIWAXS images; Figure S9: AFM images; Figures S10 and S11 and Table S1: DOS measurements; Figure S12: Chip for thermal conductivity measurement; Figures S13–S15: thermoelectric measurements; Figure S16: MIS device measurements; Figures S17–S26: NMR spectra (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Jian Liu — State Key Laboratory of Polymer Science and Technology, Changchun Institute of Applied Chemistry, Chinese Academy of Sciences, Changchun 130022, P.R. China; School of Applied Chemistry and Engineering, University of Science and Technology of China, Hefei 230026, P. R. China; orcid.org/0000-0002-6704-3895; Email: jian.liu@ciac.ac.cn

Authors

Mingyu Ma — State Key Laboratory of Polymer Science and Technology, Changchun Institute of Applied Chemistry, Chinese Academy of Sciences, Changchun 130022, P.R. China; School of Applied Chemistry and Engineering, University of Science and Technology of China, Hefei 230026, P. R. China

Gang Ye — State Key Laboratory of Polymer Science and Technology, Changchun Institute of Applied Chemistry, Chinese Academy of Sciences, Changchun 130022, P.R. China; Present Address: Key Laboratory for the Green Preparation and Application of Functional Materials, Hubei Key Laboratory of Polymer Materials, School of Materials Science and Engineering, Hubei University, Youyi Road 368, Wuhan, 430062 China; orcid.org/0000-0003-1266-0762

Soyeong Jang — King Abdullah University of Science and Technology (KAUST), Physical Science and Engineering Division (PSE), KAUST Solar Center (KSC), Thuwal 23955-6900, Saudi Arabia

Yazhuo Kuang — State Key Laboratory of Polymer Science and Technology, Changchun Institute of Applied Chemistry, Chinese Academy of Sciences, Changchun 130022, P.R. China; School of Applied Chemistry and Engineering, University of Science and Technology of China, Hefei 230026, P. R. China

Linlong Zhang — State Key Laboratory of Polymer Science and Technology, Changchun Institute of Applied Chemistry, Chinese Academy of Sciences, Changchun 130022, P.R. China; School of Applied Chemistry and Engineering, University of Science and Technology of China, Hefei 230026, P. R. China

Shuyan Shao — Institute of Molecular Aggregation Science, Tianjin University, Tianjin 300072, China

L. Jan Anton Koster — Zernike Institute for Advanced Materials, University of Groningen, NL-9747 AG Groningen, The Netherlands; orcid.org/0000-0002-6558-5295

Derya Baran — King Abdullah University of Science and Technology (KAUST), Physical Science and Engineering Division (PSE), KAUST Solar Center (KSC), Thuwal 23955-6900, Saudi Arabia; orcid.org/0000-0003-2196-8187

Complete contact information is available at:

<https://pubs.acs.org/doi/10.1021/acsenerylett.4c03590>

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

The authors are grateful for the financial support from the National Natural Science Foundation of China (No. 52273201). J.L. thanks the financial support from the Jilin Scientific and Technological Development Program (Nos. 20230402070GH and 20240101172JC) and a grant for Distinguished Young Scholars of the National Natural Science Foundation of China (Overseas). G.Y. also acknowledges the China Postdoctoral Science Foundation Funded Project (Grant No. 2022M723077). S.S. is thankful for the financial support from the National Natural Science Foundation of China (No. 62105239). The project was supported by Open Research Fund of Key Laboratory of Polymer Science and Technology, Changchun Institute of Applied Chemistry, Chinese Academy of Sciences. A portion of this work is based on the data obtained at BSRF-1W1A. The authors gratefully acknowledge the cooperation of the beamline scientists at the BSRF-1W1A beamline. The numerical calculations in this paper have been done on the supercomputing system in the Supercomputing Center of University of Science and Technology of China.

■ REFERENCES

- (1) Lüssem, B.; Riede, M.; Leo, K. Doping of organic semiconductors. *Phys. Status Solidi A* **2013**, *210* (1), 9–43.
- (2) Salzmann, I.; Heimel, G.; Duhm, S.; Oehzelt, M.; Pingel, P.; George, B. M.; Schnegg, A.; Lips, K.; Blum, R.-P.; Vollmer, A.; et al. Intermolecular Hybridization Governs Molecular Electrical Doping. *Phys. Rev. Lett.* **2012**, *108* (3), No. 035502.
- (3) Pingel, P.; Neher, D. Comprehensive picture of p-type doping of P3HT with the molecular acceptor F₄TCNQ. *Phys. Rev. B* **2013**, *87* (11), No. 115209.
- (4) Salzmann, I.; Heimel, G.; Oehzelt, M.; Winkler, S.; Koch, N. Molecular Electrical Doping of Organic Semiconductors: Fundamental Mechanisms and Emerging Dopant Design Rules. *Acc. Chem. Res.* **2016**, *49* (3), 370–378.
- (5) Wang, S.; Sun, H.; Ail, U.; Vagin, M.; Persson, P. O. Å.; Andreasen, J. W.; Thiel, W.; Berggren, M.; Crispin, X.; Fazzi, D.; et al. Thermoelectric Properties of Solution-Processed n-Doped Ladder-Type Conducting Polymers. *Adv. Mater.* **2016**, *28* (48), 10764–10771.

- (6) Liu, J.; Qiu, L.; Shao, S. Emerging electronic applications of fullerene derivatives: an era beyond OPV. *J. Mater. Chem. C* **2021**, *9* (45), 16143–16163.
- (7) Wang, H.; Yu, C. Organic Thermoelectrics: Materials Preparation, Performance Optimization, and Device Integration. *Joule* **2019**, *3* (1), 53–80.
- (8) Culebras, M.; Gómez, C. M.; Cantarero, A. Review on Polymers for Thermoelectric Applications. *Materials* **2014**, *7* (9), 6701–6732.
- (9) Tietze, M. L.; Pahner, P.; Schmidt, K.; Leo, K.; Lüssem, B. Doped Organic Semiconductors: Trap-Filling, Impurity Saturation, and Reserve Regimes. *Adv. Funct. Mater.* **2015**, *25* (18), 2701–2707.
- (10) Scheunemann, D.; Järsvall, E.; Liu, J.; Beretta, D.; Fabiano, S.; Caironi, M.; Kemerink, M.; Müller, C. Charge transport in doped conjugated polymers for organic thermoelectrics. *Chem. Phys. Rev.* **2022**, *3* (2), No. 021309.
- (11) Shao, S.; Liu, J. Molecular Insights into n-Type Organic Thermoelectrics. *CCS Chem.* **2021**, *3* (10), 2702–2716.
- (12) Scaccabarozzi, A. D.; Basu, A.; Aniés, F.; Liu, J.; Zapata-Arteaga, O.; Warren, R.; Firdaus, Y.; Nugraha, M. I.; Lin, Y.; Campoy-Quiles, M.; et al. Doping Approaches for Organic Semiconductors. *Chem. Rev.* **2022**, *122* (4), 4420–4492.
- (13) Ding, J.; Liu, Z.; Zhao, W.; Jin, W.; Xiang, L.; Wang, Z.; Zeng, Y.; Zou, Y.; Zhang, F.; Yi, Y.; et al. Selenium-Substituted Diketopyrrolopyrrole Polymer for High-Performance p-Type Organic Thermoelectric Materials. *Angew. Chem., Int. Ed.* **2019**, *58* (52), 18994–18999.
- (14) Wang, D.; Ding, J.; Ma, Y.; Xu, C.; Li, Z.; Zhang, X.; Zhao, Y.; Zhao, Y.; Di, Y.; Liu, L.; et al. Multi-heterojunctioned plastics with high thermoelectric figure of merit. *Nature* **2024**, *632* (8025), 528–535.
- (15) Bubnova, O.; Khan, Z. U.; Malti, A.; Braun, S.; Fahlman, M.; Berggren, M.; Crispin, X. Optimization of the thermoelectric figure of merit in the conducting polymer poly(3,4-ethylenedioxythiophene). *Nat. Mater.* **2011**, *10* (6), 429–433.
- (16) Fan, Z.; Li, P.; Du, D.; Ouyang, J. Significantly Enhanced Thermoelectric Properties of PEDOT:PSS Films through Sequential Post-Treatments with Common Acids and Bases. *Adv. Energy Mater.* **2017**, *7* (8), No. 1602116.
- (17) Lu, Y.; Wang, J.-Y.; Pei, J. Strategies To Enhance the Conductivity of n-Type Polymer Thermoelectric Materials. *Chem. Mater.* **2019**, *31* (17), 6412–6423.
- (18) Sun, Y.; Di, C.-A.; Xu, W.; Zhu, D. Advances in n-Type Organic Thermoelectric Materials and Devices. *Adv. Electron. Mater.* **2019**, *5* (11), No. 1800825.
- (19) Yoon, S. E.; Kang, Y.; Im, J.; Lee, J.; Lee, S. Y.; Park, J.; Gao, Y. J.; Jeon, D.; Son, J. Y.; Kim, J.; et al. Enhancing dopant diffusion for ultrahigh electrical conductivity and efficient thermoelectric conversion in conjugated polymers. *Joule* **2023**, *7* (10), 2291–2317.
- (20) Feng, K.; Guo, H.; Wang, J.; Shi, Y.; Wu, Z.; Su, M.; Zhang, X.; Son, J. H.; Woo, H. Y.; Guo, X. Cyano-Functionalized Bithiophene Imide-Based n-Type Polymer Semiconductors: Synthesis, Structure–Property Correlations, and Thermoelectric Performance. *J. Am. Chem. Soc.* **2021**, *143* (3), 1539–1552.
- (21) Tang, H.; Liang, Y.; Liu, C.; Hu, Z.; Deng, Y.; Guo, H.; Yu, Z.; Song, A.; Zhao, H.; Zhao, D.; et al. A solution-processed n-type conducting polymer with ultrahigh conductivity. *Nature* **2022**, *611* (7935), 271–277.
- (22) Dong, C.; Deng, S.; Meng, B.; Liu, J.; Wang, L. A Distannylated Monomer of a Strong Electron-Accepting Organoboron Building Block: Enabling Acceptor–Acceptor-Type Conjugated Polymers for n-Type Thermoelectric Applications. *Angew. Chem., Int. Ed.* **2021**, *60* (29), 16184–16190.
- (23) Deng, S.; Kuang, Y.; Liu, L.; Liu, X.; Liu, J.; Li, J.; Meng, B.; Di, C.-a.; Hu, J.; Liu, J. High-Performance and Ecofriendly Organic Thermoelectrics Enabled by N-Type Polythiophene Derivatives with Doping-Induced Molecular Order. *Adv. Mater.* **2024**, *36* (8), No. 2309679.
- (24) Deng, S.; Liu, J.; Meng, B.; Liu, J.; Wang, L. A Highly Conductive n-Type Polythiophene Derivative: Effect of Molecular Weight on n-Doping Behavior and Thermoelectric Performance. *ACS Appl. Mater. Interfaces* **2023**, *15* (38), 45190–45200.
- (25) Thomas, E. M.; Davidson, E. C.; Katsumata, R.; Segalman, R. A.; Chabynyc, M. L. Branched Side Chains Govern Counterion Position and Doping Mechanism in Conjugated Polythiophenes. *ACS Macro Lett.* **2018**, *7* (12), 1492–1497.
- (26) Shi, K.; Zhang, F.; Di, C.-A.; Yan, T.-W.; Zou, Y.; Zhou, X.; Zhu, D.; Wang, J.-Y.; Pei, J. Toward High Performance n-Type Thermoelectric Materials by Rational Modification of BDPPV Backbones. *J. Am. Chem. Soc.* **2015**, *137* (22), 6979–6982.
- (27) Liu, J.; Qiu, L.; Alessandri, R.; Qiu, X.; Portale, G.; Dong, J.; Talsma, W.; Ye, G.; Sengrian, A. A.; Souza, P. C. T.; et al. Enhancing Molecular n-Type Doping of Donor-Acceptor Copolymers by Tailoring Side Chains. *Adv. Mater.* **2018**, *30* (7), No. 1704630.
- (28) Kuang, Y.; Heester, S.; Shao, S.; Ye, G.; Yao, T.; Xie, Z.; Koster, L. J. A.; Liu, J. Adjusting molecular weight optimizes electronic transport of extrinsically N-type doped conjugated polymer incorporating glycolated side chains. *J. Mater. Chem. A* **2024**, *12* (8), 4866–4876.
- (29) Peng, X.; Ye, G.; Zhang, L.; Kuang, Y.; Shao, S.; Liu, J. Enhancing Charge Transport Performance of n-Doped Conjugated Polymers by Customizing Amphipathic Side Chains. *Macromolecules* **2024**, *57* (15), 7156–7164.
- (30) Liu, J.; Ye, G.; Potgieser, H. G. O.; Koopmans, M.; Sami, S.; Nugraha, M. I.; Villalva, D. R.; Sun, H.; Dong, J.; Yang, X.; et al. Amphipathic Side Chain of a Conjugated Polymer Optimizes Dopant Location toward Efficient N-Type Organic Thermoelectrics. *Adv. Mater.* **2021**, *33* (4), No. 2006694.
- (31) Liu, Q.; Bottle, S. E.; Sonar, P. Developments of Diketopyrrolopyrrole-Dye-Based Organic Semiconductors for a Wide Range of Applications in Electronics. *Adv. Mater.* **2020**, *32* (4), No. 1903882.
- (32) Yu, Y.; Wu, Y.; Zhang, A.; Li, C.; Tang, Z.; Ma, W.; Wu, Y.; Li, W. Diketopyrrolopyrrole Polymers with Thienyl and Thiazolyl Linkers for Application in Field-Effect Transistors and Polymer Solar Cells. *ACS Appl. Mater. Interfaces* **2016**, *8* (44), 30328–30335.
- (33) Yu, H.; Kim, H. N.; Song, I.; Ha, Y. H.; Ahn, H.; Oh, J. H.; Kim, Y.-H. Effect of alkyl chain spacer on charge transport in n-type dominant polymer semiconductors with a diketopyrrolopyrrole-thiophene-bithiazole acceptor-donor-acceptor unit. *J. Mater. Chem. C* **2017**, *5* (14), 3616–3622.
- (34) Chávez, P.; Bulut, I.; Fall, S.; Ibraikulov, O. A.; Chochos, C. L.; Bartringer, J.; Heiser, T.; Lévesque, P.; Leclerc, N. An Electron-Transporting Thiazole-Based Polymer Synthesized Through Direct (Hetero)Arylation Polymerization. *Molecules* **2018**, *23* (6), 1270.
- (35) Jung, I. H.; Hong, C. T.; Lee, U.-H.; Kang, Y. H.; Jang, K.-S.; Cho, S. Y. High Thermoelectric Power Factor of a Diketopyrrolopyrrole-Based Low Bandgap Polymer via Finely Tuned Doping Engineering. *Sci. Rep.* **2017**, *7* (1), 44704.
- (36) Perry, E. E.; Chiu, C.-Y.; Moudgil, K.; Schlitz, R. A.; Takacs, C. J.; O'Hara, K. A.; Labram, J. G.; Claudell, A. M.; Sherman, J. B.; Barlow, S.; et al. High Conductivity in a Nonplanar n-Doped Ambipolar Semiconducting Polymer. *Chem. Mater.* **2017**, *29* (22), 9742–9750.
- (37) Yang, C.-Y.; Jin, W.-L.; Wang, J.; Ding, Y.-F.; Nong, S.; Shi, K.; Lu, Y.; Dai, Y.-Z.; Zhuang, F.-D.; Lei, T.; et al. Enhancing the n-Type Conductivity and Thermoelectric Performance of Donor–Acceptor Copolymers through Donor Engineering. *Adv. Mater.* **2018**, *30* (43), No. 1802850.
- (38) Yan, X.; Xiong, M.; Li, J.-T.; Zhang, S.; Ahmad, Z.; Lu, Y.; Wang, Z.-Y.; Yao, Z.-F.; Wang, J.-Y.; Gu, X.; et al. Pyrazine-Flanked Diketopyrrolopyrrole (DPP): A New Polymer Building Block for High-Performance n-Type Organic Thermoelectrics. *J. Am. Chem. Soc.* **2019**, *141* (51), 20215–20221.
- (39) Shi, Y.; Zhang, X.; Du, T.; Han, Y.; Deng, Y.; Geng, Y. A High-Performance n-Type Thermoelectric Polymer from C–H/C–H Oxidative Direct Arylation Polycondensation. *Angew. Chem., Int. Ed.* **2023**, *62* (20), No. e202219262.

- (40) Cai, H.; Tang, H.; Wang, T.; Xu, C.; Xie, J.; Fu, M.; Luo, X.; Hu, Z.; Zhang, Y.; Deng, Y.; et al. An n-Type Open-Shell Conjugated Polymer with High-Spin Ground-State and High Intrinsic Electrical Conductivity. *Angew. Chem., Int. Ed.* **2024**, 63 (25), No. e202402375.
- (41) Tu, L.; Wang, J.; Wu, Z.; Li, J.; Yang, W.; Liu, B.; Wu, S.; Xia, X.; Wang, Y.; Woo, H. Y.; et al. Cyano-Functionalized Pyrazine: A Structurally Simple and Easily Accessible Electron-Deficient Building Block for n-Type Organic Thermoelectric Polymers. *Angew. Chem., Int. Ed.* **2024**, 63 (11), No. e202319658.
- (42) Glaudell, A. M.; Cochran, J. E.; Patel, S. N.; Chabiny, M. L. Impact of the Doping Method on Conductivity and Thermopower in Semiconducting Polythiophenes. *Adv. Energy Mater.* **2015**, 5 (4), No. 1401072.
- (43) Kang, S. D.; Snyder, G. J. Charge-transport model for conducting polymers. *Nat. Mater.* **2017**, 16 (2), 252–257.
- (44) Abdalla, H.; Zuo, G.; Kemerink, M. Range and energetics of charge hopping in organic semiconductors. *Phys. Rev. B* **2017**, 96 (24), No. 241202.
- (45) Gregory, S. A.; Hanus, R.; Atassi, A.; Rinehart, J. M.; Wooding, J. P.; Menon, A. K.; Losego, M. D.; Snyder, G. J.; Yee, S. K. Quantifying charge carrier localization in chemically doped semiconducting polymers. *Nat. Mater.* **2021**, 20 (10), 1414–1421.
- (46) Liu, J.; Ye, G.; Zee, B. v. d.; Dong, J.; Qiu, X.; Liu, Y.; Portale, G.; Chiechi, R. C.; Koster, L. J. A. N-Type Organic Thermoelectrics of Donor–Acceptor Copolymers: Improved Power Factor by Molecular Tailoring of the Density of States. *Adv. Mater.* **2018**, 30 (44), No. 1804290.
- (47) Wang, X.-Y.; Yu, Z.-D.; Lu, Y.; Yao, Z.-F.; Zhou, Y.-Y.; Pan, C.-K.; Liu, Y.; Wang, Z.-Y.; Ding, Y.-F.; Wang, J.-Y.; et al. Density of States Engineering of n-Doped Conjugated Polymers for High Charge Transport Performances. *Adv. Mater.* **2023**, 35 (21), No. 2300634.
- (48) Wang, X.-Y.; Liu, Y.; Wang, Z.-Y.; Lu, Y.; Yao, Z.-F.; Ding, Y.-F.; Yu, Z.-D.; Wang, J.-Y.; Pei, J. Revealing the effect of oligo(ethylene glycol) side chains on n-doping process in FBDPPV-based polymers. *J. Polym. Sci.* **2022**, 60 (3), 538–547.
- (49) Chen, X.-X.; Li, J.-T.; Fang, Y.-H.; Deng, X.-Y.; Wang, X.-Q.; Liu, G.; Wang, Y.; Gu, X.; Jiang, S.-D.; Lei, T. High-mobility semiconducting polymers with different spin ground states. *Nat. Commun.* **2022**, 13 (1), 2258.
- (50) Yanagi, K.; Kanda, S.; Oshima, Y.; Kitamura, Y.; Kawai, H.; Yamamoto, T.; Takenobu, T.; Nakai, Y.; Maniwa, Y. Tuning of the Thermoelectric Properties of One-Dimensional Material Networks by Electric Double Layer Techniques Using Ionic Liquids. *Nano Lett.* **2014**, 14 (11), 6437–6442.
- (51) Hulea, I. N.; Brom, H. B.; Houtepen, A. J.; Vanmaekelbergh, D.; Kelly, J. J.; Meulenkaamp, E. A. Wide Energy-Window View on the Density of States and Hole Mobility in Poly(p-Phenylene Vinylene). *Phys. Rev. Lett.* **2004**, 93 (16), No. 166601.
- (52) Huang, L.; Eedugurala, N.; Benasco, A.; Zhang, S.; Mayer, K. S.; Adams, D. J.; Fowler, B.; Lockart, M. M.; Saghayezian, M.; Tahir, H.; et al. Open-Shell Donor–Acceptor Conjugated Polymers with High Electrical Conductivity. *Adv. Funct. Mater.* **2020**, 30 (24), No. 1909805.
- (53) Yang, X.; Ye, G.; Liu, J.; Chiechi, R. C.; Koster, L. J. A. Carrier–Carrier Repulsion Limits the Conductivity of N-Doped Organic Semiconductors. *Adv. Mater.* **2024**, 36 (44), No. 2404397.
- (54) Linseis, V.; Völklein, F.; Reith, H.; Nielsch, K.; Woias, P. Advanced platform for the in-plane ZT measurement of thin films. *Rev. Sci. Instrum.* **2018**, 89 (1), No. 015110.
- (55) Liu, J.; Craighero, M.; Gupta, V. K.; Scheunemann, D.; Paletti, S. H. K.; Järsvall, E.; Kim, Y.; Xu, K.; Reparaz, J. S.; Koster, L. J. A.; et al. Electrically Programmed Doping Gradients Optimize the Thermoelectric Power Factor of a Conjugated Polymer. *Adv. Funct. Mater.* **2024**, 34 (18), No. 2312549.
- (56) Kang, K.; Watanabe, S.; Broch, K.; Sepe, A.; Brown, A.; Nasrallah, I.; Nikolka, M.; Fei, Z.; Heeney, M.; Matsumoto, D.; et al. 2D coherent charge transport in highly ordered conducting polymers doped by solid state diffusion. *Nat. Mater.* **2016**, 15 (8), 896–902.
- (57) Levy, M.; Sarachik, M. P. Measurement of the Hall coefficient using van der Pauw method without magnetic field reversal. *Rev. Sci. Instrum.* **1989**, 60 (7), 1342–1343.
- (58) Yi, H. T.; Gartstein, Y. N.; Podzorov, V. Charge carrier coherence and Hall effect in organic semiconductors. *Sci. Rep.* **2016**, 6 (1), 23650.
- (59) Döhler, G. H. Conductivity, thermopower, and statistical shift in amorphous semiconductors. *Phys. Rev. B* **1979**, 19 (4), 2083–2091.
- (60) Arkhipov, V. I.; Emelianova, E. V.; Heremans, P.; Bässler, H. Analytic model of carrier mobility in doped disordered organic semiconductors. *Phys. Rev. B* **2005**, 72 (23), No. 235202.